A070 v0.2.1 | Strict Synthetic Blind Patch

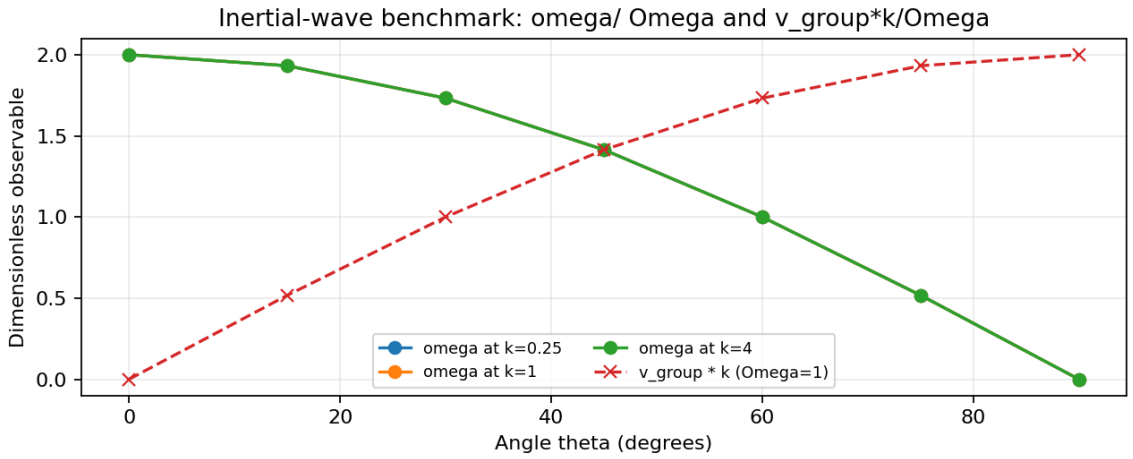
SST Absolute Time / Swirl Clock / Torsional-Light Candidate. Dated 18 October 2026. DEVELOPMENT / NOT FRAMEWORK CERTIFIED / NO SOURCE-NATIVE PKLSA DATA.

Purpose and decisive audit findings

The previous pilot used a 50-degree wavevector angle and set Omega = v_swirl/r_c with canonical SST constants. Those choices were illustrative, not physically derived. The inferred SI group speed was therefore a reference-scale consequence of input parameters, not an independently predicted light speed. This patch avoids both selection mechanisms in the BLIND inference.

Pre-registered, dimensionless sweep

Control	Frozen inputs / treatment
Angle	0, 15, 30, 45, 60, 75, 90 degrees
Wavenumber k	0.25, 0.5, 1, 2, 4 in dimensionless units
Observed data	Complex time series for each (angle, k); unknown masked identities
Blind decision	Fit frequency patterns with frozen relative residual 0.035; report unknown when no fit
Forbidden calibrations	No c, v_swirl, r_c, photon target frequency, or post hoc A/f tuning

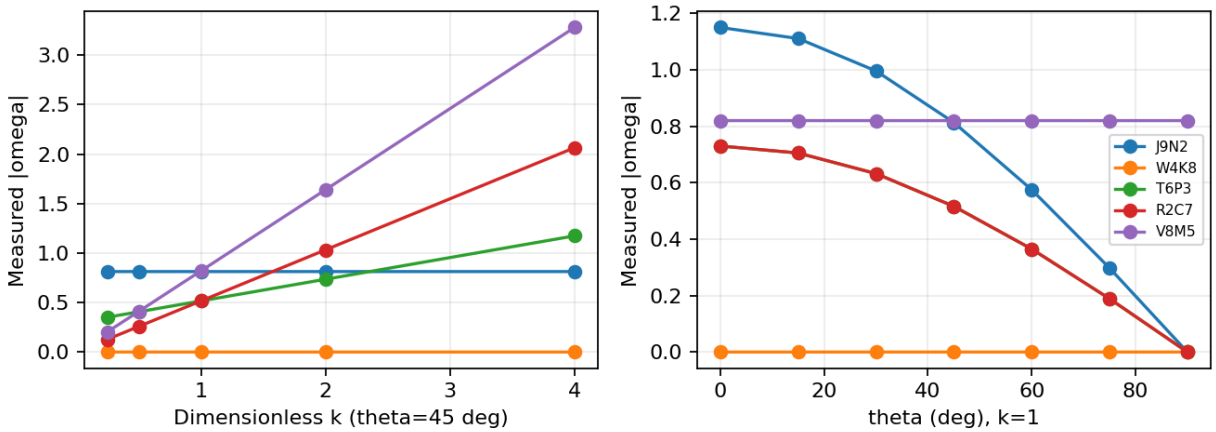


Analytic reference only: in a uniformly rotating Euler fluid, |omega| = 2 Omega |cos(theta)| and |v_group| = 2 Omega |sin(theta)| / k. The theta = 90 deg endpoint has zero eigenfrequency and must not be claimed as a propagating photon. The group-velocity vector is perpendicular to k.

Blind mechanism-classification test

The BLIND archive contains five masked synthetic families. Their generating mechanisms, numerical amplitude settings and masked-ID mapping are stored only in a separate REVEAL archive. Source-native PKLSA modes are NOT present. Correctly detecting a synthetic wave-like positive control is not evidence that torsional photons exist in an inviscid fluid.

Blinded observations only (no generator or truth used)



Frozen classifications (labels here are model decisions, not ground truth)

Masked ID	Blind mode decision	Relative RMS
J9N2	inertial_reference	0.00001
W4K8	stationary	0.00000
T6P3	unknown	0.27889
R2C7	advective_reference	0.00001
V8M5	synthetic_isotropic_linear	0.00001

Frozen outcome SHA-256: f568aedaec69dfa32b1848e796e765c53bee7cdc0ef8ea77bfd612673faa1560

What this does and does not prove

The masked run verifies that its classification workflow can distinguish predeclared dispersive/advective/zero-frequency patterns and can reject a nonmatching family. It does NOT determine the existence, speed, helicity, energy flux or local-clock coupling of a physical finite-core torsion pulse.

Physics closure and gate ledger

Independent reference dynamics

The incompressible inviscid Euler equations in absolute time t are: ∂u/∂t + (u·∇)u = −∇p/rho_f, with ∇·u = 0. Small, solenoidal perturbations in the unforced quiescent uniform fluid cannot form a propagating shear/torsion wave at linear order. In uniform axial advection, lab-frame frequency changes by k·U without new intrinsic wave propagation. Uniform rigid rotation permits anisotropic dispersive inertial waves, but that is not a finite-core vortex photon.

SST measurement distinctions

Absolute time t and Euclidean background distance x are independent evolution and spatial coordinates. Atomic Swirl Clocks accumulate tau_i, where f_i = d tau_i/dt must arise from native dynamics. Operational distance l_op must also be independently derived, with local directional length factor A = d l_op/dx. The identity c_local=(A/f)c_abs does not supply a physics law for f, A or c_abs.

Gate decisions

Gate	Status	Restriction
T0-T2	REFERENCE ONLY	Euler null/advection/inertial analytic controls
T3	BLIND FROZEN	Five masked synthetic classes; truth still separately sealed
T4	PASS CODE CHECK	No canonical speed/radius/light speed in blind engine
T5	UNRESOLVED	No source-native finite-core PKLSA torsional eigenmode
T6-T7	UNRESOLVED	Clock, rod, moving-observer closure missing
T8	NOT RUN	No predictions against light bending, GPS, SN/BAO/CMB
T9	NOT CERTIFIED	Not instantiated with framework v1.0.7

Independent physical blind next step

Obtain finite-core 3D PKLSA fields and provenance from a source custodian; pre-freeze the topology, perturbation, angle/k grids, spatial and temporal refinements, and classification thresholds. Keep sample identity and validation truth with a separate custodian. Do not unblind and retune to attain c or cosmological observables.

Reproduction, limitations and references

Reproduction

From the extracted BLIND project, run python run_all.py SELFTEST, python run_all.py BLIND, then python run_all.py VERIFY. The BLIND output is written to results/BLIND_CLASSIFICATION.json with the immutable digest results/FROZEN_SHA256.txt. Running the analysis does not need or open the REVEAL archive.

After independently locking the result, verify the REVEAL archive digest against preregistration/REVEAL_ARCHIVE_SHA256.txt, then extract it and run python evaluate.py [blind-project-path]. The evaluator refuses to score if the frozen blind digest no longer matches. The BLIND archive contains the science contract, data checksum manifest, pipeline and tests; the REVEAL archive contains the separate truth and synthetic generator settings.

Critical falsification boundaries

Torsion is an angular or azimuthal perturbation, not a transverse filament displacement. The observed isotropic synthetic positive control is an imposed wave, not a result of Euler/PKLSA dynamics. A rotating inertial wave is not equivalent to light. No claims of gravity, cosmic acceleration, universal photon speed or torsion-wave energy propagation are certified by this study.

References (bibliography entries also in REPORT.tex)

G. K. Batchelor (1967), An Introduction to Fluid Dynamics, Cambridge University Press. DOI: 10.1017/CBO9780511800955.

H. P. Greenspan (1968), The Theory of Rotating Fluids, Cambridge University Press. ISBN 9780521043885.

P. G. Saffman (1992), Vortex Dynamics, Cambridge University Press. DOI: 10.1017/CBO9780511624063.

N. Ashby (2003), Relativity in the Global Positioning System, Living Reviews in Relativity 6, 1. DOI: 10.12942/lrr-2003-1.

This release preserves A070 v0.2.0 unchanged and is an honest synthetic blinding patch. A qualified native PKLSA falsifier remains a subsequent study.